Multimodal Remote Sensing Image Clustering With Multiscale Spectral–Spatial Anchor Graphs

Xinxin Wang, Yongshan Zhang, *Member, IEEE*, and Yicong Zhou, *Senior Member, IEEE*

***Abstract*— Existing multiview clustering methods have achieved remarkable success for general images (GIs), but still have many limitations for clustering multimodal remote sensing images (RSIs). For example, these methods are sensitive to noise and spectral variability, ignore the diverse spatial structure information across modalities, or are computationally prohibitive for large-scale RSIs, thereby limiting their applications. This article proposes a multiscale spectral–spatial anchor graph fusion (MSSAGF) method for multimodal RSI clustering. MSSAGF develops a superpixel-based nonlinear neighborhood recovery strategy to reduce noise while enhancing spatial smoothness in multimodal RSIs. Using spatial-aware anchors to extract local spatial information for each modality, MSSAGF introduces multiscale local spectral–spatial anchor graphs to capture nonlinear correlations between the pixels and their corresponding local regions. A small number of anchors effectively reduces graph construction and partitioning costs, making the time complexity of MSSAGF nearly linear. This ensures that it is computationally feasible for large-scale RSIs. Finally, MSSAGF develops an adaptive fusion mechanism to fuse multiscale local anchor graphs into a unified global anchor graph, integrating complementary information across multiple modalities while directly obtaining the final clustering results. The experimental results on three multimodal RSI datasets demonstrate the superiority of our proposed method over state-of-the-art methods. Our code is publicly available at https://github.com/W-Xinxin/MSSAGF.**

***Index Terms*— Anchor graph, multimodal clustering, multiscale fusion, remote sensing images (RSIs), superpixel segmentation.**

I. Introduction

NOWADAYS, remote sensing technology has entered the fast lane, and a variety of devices have been deployed to observe the Earth's surface. The increasing diversity of remote sensing image (RSI) data brings new opportunities to the Earth observation community. Compared with single-modal data, multimodal data provides more comprehensive information for Earth observation. Formally, multimodal RSI data consists of at least two different modalities that record consensus land cover and offer complementary information for each other. For instance, digital surface model (DSM) images consist of height information to present the ground elevation of different objects [1]. Hyperspectral (HS) images contain hundreds of spectral bands that provide informative reflected radiation signals for land-cover classification [2]. The joint utilization of DSM images and HS images offers a feasible solution for identifying the asphalt on the roads or the roofs of buildings. Accordingly, joint land-cover recognition of multimodal RSI data has become an emerging cross-disciplinary field between remote sensing and machine learning communities [3].

In the past decade, supervised multimodal RSI classification has been extensively explored and evolved from the classic method to the large model [4]. Some early approaches concatenated multiple modalities and then traditional classification approaches can be applied [5], [6]. Moreover, [7] developed a latent subspace learning method based on concatenated modalities to predict the label information of pixels. With powerful representation learning capabilities, deep learning-based methods focus on extracting semantic information from multiple modalities for RSI classification. For example, [8] proposed a deep hierarchical vision transformer, and [9] proposed a collaborative contrastive learning method. Recently, [10] proposed a universal multimodal large language model for multimodal RSI classification tasks. However, these methods, particularly deep learning approaches, heavily rely on extensive training data. This necessitates pixel annotations of RSIs, resulting in labor-intensive and time-consuming efforts.

In contrast, unsupervised multimodal RSI clustering explores the similarities among pixels and correlations among modalities to achieve unsupervised recognition. This can release the dependency of massive pixel annotations. In real-world applications, it is a challenging task due to the inherent spectral variability and significant differences between diverse modalities [11]. Recent studies have investigated multimodal RSI clustering. The early methods are all deep learning approaches, requiring complex optimization procedures [12], [13], [14]. Other works [15], [16], [17] employ anchor graphs to handle large-scale RSI datasets. However, these methods utilize only graph structures with the same scale across modalities, leading to suboptimal performance. In addition to the remote sensing community, the machine learning community focuses on clustering multimodal data earlier, for example, general images (GIs) data. Typically, these methods are referred to as multiview clustering methods. In recent years, the research of graph-based multiview clustering (GMVC) has

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achieved significant progress, resulting in a wide range of methodologies. For example, distance measure-based GMVC methods [18], subspace-based GMVC methods [19], kernel-based GMVC methods [20], and bipartite graph-based GMVC methods [21]. Besides, deep learning-based multiview clustering has achieved huge success [22], [23], [24]. By treating multimodal RSI as multiple views, one can rapidly apply multiview clustering methods for clustering tasks.

However, these aforementioned methods still suffer from several limitations. First, the multimodal RSI data usually suffer from noise and data corruption due to sensor offset and electromagnetic interference between load components [2]. The spectral variability problem aggravates this significant challenge [25]. Consequently, directly applying existing methods to raw data is insufficient to effectively model such complex data. Second, multimodal RSIs typically contain rich spatial information and may exhibit different spatial textures across different modalities [26]. Existing multimodal methods leverage only the spatial information through the graph structures of the same size, ignoring the discrepancies among modalities and thereby limiting clustering effectiveness. Third, most methods require quadratic and cubic time complexity for constructing similarity graphs and performing eigenvalue decomposition. This is computationally prohibitive for large-scale multimodal RSI data [27]. Moreover, existing methods ignore the interpretability of clustering results. This is crucial for land-cover recognition in Earth observation.

To address these limitations, we propose a multiscale spectral–spatial anchor graph fusion (MSSAGF) method for multimodal RSI clustering. To the best of our knowledge, this is the first work to implement the fusion of anchor graphs at different scales, fully exploiting the spatial structure information of multimodal RSIs. The main contributions of this article are summarized as follows.

1) We propose a scalable multimodal RSI clustering method based on MSSAGF. Our approach begins with superpixel-based nonlinear neighborhood recovery to enhance RSI data, followed by multiscale spectral–spatial anchor graph construction. Subsequently, using multiscale local anchor graphs as input, MSSAGF develops an adaptive fusion mechanism that integrates global anchor graph learning and graph partitioning process into a unified framework, which directly generates clustering indicators.
2) We propose a superpixel-based neighborhood recovery strategy to reduce noise and enhance spatial smoothness in RSIs. This strategy performs independent superpixel segmentation for each modality, allowing for the full exploitation of view-specific texture information through manifold learning in the superpixels.
3) Using view-specific superpixel segmentation results and recovered RSI data, we generate a diverse number of spatial-aware anchors to fully capture the local spatial information of each modality. Next, we construct multiscale local spectral–spatial anchor graphs for each modality to establish nonlinear correlations between pixels and local homogeneous regions. Compared to full graph structures, anchor graphs effectively reduce computational costs.
4) We devise an alternating optimization algorithm to solve the proposed formulation. Extensive experiments verify the effectiveness of our method.

The rest of this article is structured as follows. Section II introduces the related work. The proposed method is elaborated in Section III. Section IV demonstrates extensive experimental results and analyses. Finally, Section V concludes this article.

II. Related Work

A. Anchor GMVC

Anchor graph strategy has achieved attractive attention, due to its scalability for large-scale datasets [28]. Kang et al. [29] proposed an anchor-based subspace representation learning approach to achieve linear time complexity for multiview clustering. Moreover, [30] developed a structured anchor graph learning method to obtain clustering labels based on connected components. Rather than using fixed anchors, [21] introduced a dynamic anchor learning strategy to enhance the flexibility of the corresponding anchor graph. Recently, tensor learning techniques have been implemented to explore the high-order correlations for anchor graphs among multiple views [31]. A fuzzy learning strategy has been incorporated into the anchor graph, addressing the issues of label uncertainty [32]. Furthermore, a weighting-based noisy feature filter was proposed to address the anchor shift problem within anchor graph construction [33]. These methods have achieved significant success in clustering GI datasets. However, they often suffer from performance degradation when directly applied to RSI datasets. This is because they ignore the spatial information of pixels in RSIs.

B. Graph-Based Clustering for RSIs

Graph-based clustering leverages the advantage of geometric structure to present the relationships between samples, demonstrating promising applications for unsupervised recognition of remote sensing data. Zhang et al. [34] incorporated spectral and spatial information into the sparse subspace clustering algorithm to obtain an accurate geometric structure, to which spectral clustering is applied to yield final clustering results. Zhao et al. [35] leveraged superpixels to capture the spatial interdependencies of neighboring pixels and proposed superpixel-level graph clustering. Cai et al. [36] employed an autoencoder to implement superpixel-level subspace network learning. To enhance computational efficiency, [37] proposed to achieve fast spectral clustering using a bipartite graph for HS image data. Wang et al. [38] proposed a nonnegative relaxation spectral embedding to accelerate graph partitioning. However, these methods only cluster a single data source (modality), making them ineffective for handling multiple data modalities.

C. Graph-Based Clustering for Multimodal RSIs

To explore comprehensive information from multiple perspectives, [39] partitioned a single RSI into different views

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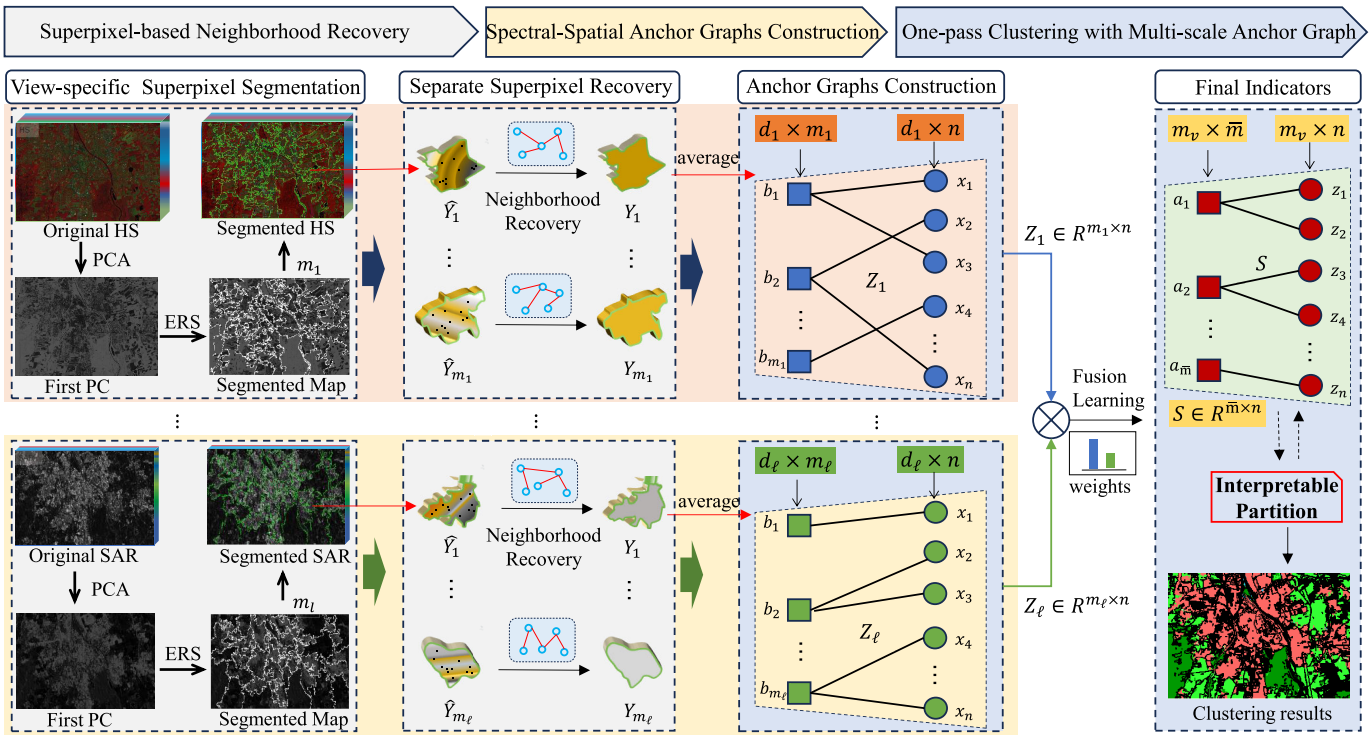


Fig. 1. Overall flowchart of the proposed MSSAGF for multimodal RSI clustering. It performs superpixel segmentation separately on each modality of RSIs to obtain the segmented RSI. Based on generated superpixels, nonlinear neighborhood recovery is performed on each superpixel, and then the average spectrum of each superpixel is regarded as a spatial-aware anchor, resulting different number of anchors for each modality. Given the recovered pixels and spatial-aware anchors of each modality, different scales of spectral–spatial anchor graphs are constructed, respectively. Taking anchor graphs as newly structural representation data, unified global anchor graph learning and interpretable partition are simultaneously conducted to directly output clustering results.

based on various criteria and performed multiview subspace clustering. Huang et al. [40] utilized a spatial feature extractor to create a second view and proposed hypergraph regularized multiview subspace clustering. However, these methods still primarily work with single-modal RSI data and neglect the discrepancies between multiple modalities. Moreover, they are computationally prohibitive when dealing with large-scale RSI datasets due to their full graph structure. Recently, some works have employed anchor graphs to achieve scalable multimodal RSI clustering. For example, [15] developed a joint clustering model for HS and LiDAR data based on multiview kernel subspace clustering. Zhang et al. [16] introduced a dual-graph learning method for multimodal RSI clustering. Zhang et al. [17] proposed a fast clustering method that leverages fuzzy labels of anchors. These methods have achieved promising performance. However, they only consider graph structures at the same scale across modalities, while ignoring the spectral–spatial structural specificity of each modality.

III. Proposed Method

In this section, we first elaborate on the proposed multimodal RSI clustering method, consisting of three steps: 1) superpixel-based nonlinear neighborhood recovery; 2) multiscale spectral–spatial anchor graph construction; and 3) one-pass clustering with MSSAGF. Then, we describe the optimization procedure in detail, followed by complexity analysis.

A. Overview

The whole flowchart of the proposed MSSAGF is displayed in Fig. 1. **First**, to address the noise-sensitive and spectral variability problem, each RSI is divided into multiple homogeneous regions independently by using the entropy rate superpixel (ERS) segmentation technique [41]. Subsequently, pixels in each homogeneous region are recovered from the corresponding geometric neighborhood of the pixels in the superpixel block. This process can strengthen the spatial smoothness of the RSI. **Second**, to leverage the spatial information, spatial-aware anchors are generated from each recovered superpixel, and spectral–spatial anchor graphs are constructed to display the group structure of each modality, respectively. Unlike existing methods that select the same number of anchors for each modality to facilitate subsequent anchor graph fusion, we determine the anchor number based on the results of homogeneous region segmentation for each modality. This approach reflects the spectral–spatial specificity of each modality and results in anchor graphs at different scales, considering modality-specific clustering structure optimally. **Third**, to tackle with computational expenditure problem of large-scale RSIs, we propose an adaptive fusion mechanism to fuse multiscale local anchor graphs into a unified global anchor graph. This model treats the anchor graph as an anchor-based representation matrix, to newly learn the global group structure embedded in multiple local anchor graphs. Furthermore, we incorporate a partition process to make the proposed model exist interpretability. This can directly obtain the clustering results without extra post-processing. Compared with existing multimodal clustering methods, the proposed MSSAGF can effectively explore the global group structure by considering the optimal scale spectral–spatial anchor graph for each modality. Notably,

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it efficiently tackles the fusion problem of different scales of anchor graphs.

Let us denote multimodal RSIs by $\{\mathcal{X}^v \in \mathbb{R}^{h\times w\times d_v}\}_{v=1}^{\ell}$, where h, w, d_v, and ℓ are the numbers of height, width, spectral bands, and modalities, respectively. In the RSI clustering task, our goal is to recognize the land-cover category for each pixel.

B. Superpixel-Based Nonlinear Neighborhood Recovery

Due to the unstable observation condition, noise interference is ubiquitous in RSIs, severely leading to performance degradation in subsequent learning tasks. To this end, many traditional spectral–spatial feature learning methods tend to enhance the spatial information via window filter techniques [34], [42], but the spatial structures may not be effectively extracted because the rectangular window is overly strict in nature and may cause unexpected spectral distortion. To respond to this problem, we use the ERS segmentation technique to extract the homogeneous objects separately on each modality [43], and then the geometric neighborhood recovery model is developed to alleviate noise and recover effective spatial information for each homogeneous region.

1) View-Specific Homogeneous Region Generation: As a typical unsupervised technique for recognizing homogeneous regions, the ERS segmentation approach was originally designed for red, green, blue (RGB) image segmentation. ERS requires grayscale input, necessitating the conversion of RGB images to grayscale before segmentation. However, since RSI data contain multiple spectral bands, a representative component must first be selected for segmentation. Principal component analysis (PCA) is a common dimension reduction technique used as a preprocessing step for ERS segmentation in comparative methods [35], [36]. For a fair comparison, we also adopt PCA to obtain the first principal component as the input for ERS. Of course, other dimension reduction methods can replace PCA. The purpose of using PCA in this article is to capture the essential information from RSI data. Although using dimension reduction approaches to obtain the first principal component may result in some loss of spatial information, recent studies [11], [16] have shown that PCA is effective and sufficient for extracting essential information for generating homogeneous regions.

We denote the first principal component of each RSI extracted by PCA as $\{I_f^v \in \mathbb{R}^{h\times w}\}_{v=1}^{\ell}$. Then, ERS is separately executed on each I_f^v to generate homogeneous superpixel blocks as follows:

$$I_f^v = \bigcup_{p=1}^{m_v} \mathcal{P}_p^v, \quad \text{s.t. } \mathcal{P}_p^v \cap \mathcal{P}_g^v = \emptyset, \ (p \neq g) \tag{1}$$

where m_v is the number of superpixels and $\mathcal{P}_p^v$ indicates the pth superpixel on the vth modality. Each superpixel indicates a group of locally confined pixels that share similar characteristics. It is worth noting that the number of superpixels of each modality is different, because each modality records different physical signals of land cover, resulting in diverse spatial contexts. In this article, we follow [44] and adaptively set the superpixel number according to the spatial textures of each modality.



(a) (b) (c)

Fig. 2. False color composites. (a) Raw. (b) Superpixel-based locally reconstructed image [25]. (c) Superpixel-based neighborhood recovered image.

2) Superpixel-by-Superpixel Recovery: To remove noise and alleviate the spectral variability problem in each superpixel block, the superpixel-based local reconstruction methods have been developed for RSI denoising [25], [45]. However, these types of methods that essentially perform local linear smoothing on spatial neighbors or spectral neighbors cannot deal with features on nonlinear manifolds. In this article, we suppose that the original features of all pixels in a superpixel block should be close to the observed features, and they distribute in nonlinear geometric manifolds in nature. Assume the manifold structure of all pixels in a superpixel is $\mathcal{W} \in \mathbb{R}^{q\times q}$ (q is the number of pixels in the superpixel). Then, we recover the original features of all pixels as follows:

$$\min_{\mathbf{Y}} \left\|\hat{\mathbf{Y}} - \mathbf{Y}\right\|_F^2 + \lambda \operatorname{Tr}\left(\mathbf{Y}\mathcal{L}\mathbf{Y}^T\right) \tag{2}$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{d_v\times q}$ indicates the observed pixel matrix of a superpixel with each pixel as a column vector and $\mathbf{Y} \in \mathbb{R}^{d_v\times q}$ is the recovered pixel matrix of the superpixel. λ is a parameter to balance the manifold constraint and distance deviation. $\mathcal{L} = \mathcal{D} - \mathcal{W}$ is a Laplacian matrix and $\mathcal{D}$ is a degree matrix with diagonal elements $\mathbf{D}_{i,i} = \sum_{j=1}^{q} \mathcal{W}_{i,j}$.

Based on that, we replace the observed pixels with the newly produced pixels and repeat the above operations on all superpixels till to recover the whole region of an RSI. For multimodal RSIs, we repeat the above operations on each modality. Therefore, the recovered RSIs are modality-oriented and region-aware new representations.

To demonstrate the effectiveness of the proposed superpixel-based neighborhood recovery, three false color composites of an RSI are displayed in Fig. 2. The raw, locally reconstructed, and recovered data are shown from left to right. We can observe that the noisy points in the raw image, for example, unordered white and black noisy points on top and bottom, are eliminated by both our proposed method and the local reconstruction method. Furthermore, our proposed approach preserves the boundaries between different homogeneous regions better in comparison with the local reconstruction method.

C. Multiscale Spectral–Spatial Anchor Graph Construction

Graph structure can capture relationships among samples to disclose the underlying data structure for clustering tasks. Compared to an $n \times n$ full graph, an $n \times m$ anchor graph has a more compact structure since $m \ll n$. An anchor graph has shown its capacity in dealing with large-scale data, due to lower time complexity in constructing and lower space costs in

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storing. To efficiently tackle multimodal large-scale RSI clustering in this article, we use anchor graphs to characterize both the relationships between the pixels of each RSI and the cluster structure on each modality. Several forms of anchor graph construction on multiview data have been presented [21], [27]. However, these existing methods have two main issues: 1) the existing anchor selection strategies in the machine learning community, for example, random anchors [37] or k-means anchors [30], cannot consider the spatial information of the RSI, making them unsuitable for spatial interdependency of land cover in practice and 2) they set the same number of anchors and construct an anchor graph of the same scale on each modality, ignoring the data structure distinctions between multiple modalities of RSIs.

1) Spatial-Aware Anchors Generation: To ensure that the anchor graphs sufficiently reflect the pixel affiliation to local homogeneous regions, we intend to generate spatial-aware anchors $\mathbf{B}^v = \{\mathbf{b}_1, \ldots, \mathbf{b}_l, \ldots, \mathbf{b}_{m_v}\} \in \mathbb{R}^{d_v \times m_v}$ based on the spatial texture information for each modality, allowing the generated anchors to represent the optimal spatial characteristics of each modality. Specifically, we adaptively segment the RSI of the vth modality into m_v superpixels, as introduced in Section III-B1. We then generate the mean spectrum from the lth superpixel as an anchor $\mathbf{b}_l$ to preserve the spectral–spatial discrimination of the generated anchors, where l ranges from 1 to m_v. This approach allows the generated superpixel-based anchors to capture spatial information accurately.

2) Anchor Graph Construction: Using the above spatial-aware anchors, we then construct a spectral–spatial anchor graph for each modality, respectively. The anchor graph $\mathbf{Z}^v \in \mathbb{R}^{n \times m_v}$, a cross similarity matrix between n data points and m_v anchors, can be constructed in a variety of methods, for example, [21], [37]. The majority are based on the similarity between two samples, which is inversely proportional to the distance of them. We use the approach of [46] to construct anchor graphs. Specifically, the similarity probability $\mathbf{z}^v_{i,j}$ between the ith data point $\mathbf{x}_i$ and one of the nearest c anchor points $\mathbf{b}_j$ can be computed as follows:

$$\mathbf{z}^v_{ij} = \frac{\overline{\mathbf{d}}_{i,c+1} - \overline{\mathbf{d}}_{ij}}{c\overline{\mathbf{d}}_{i,c+1} - \sum_{j=1}^{c} \overline{\mathbf{d}}_{ij}} \tag{3}$$

where $\overline{\mathbf{d}}_i \in \mathbb{R}^{m_v \times 1}$ denotes the Euclidean distance vector between the ith data point and m_v anchors in ascending order. c indicates the nearest c anchor points are assigned similarity probability and the rest are 0. This article utilizes each pixel to construct an anchor graph. However, it is very difficult to perform pixel-wise calculations directly on a 3-D RSI cube. Applying a reshape operation is a common approach to facilitate subsequent computations and it is widely used in the competing algorithms [37], [38]. Specifically, we reshape each original 3-D cube into a 2-D matrix $\mathbf{X}^v \in \mathbb{R}^{d_v \times n}$ ($n = h \times w$) and then use the 2-D matrix along with the anchor matrix to compute Euclidean distance through matrix multiplication. Spatial–spectral anchors reflect the spatial–spectral characteristic of a local region, allowing the constructed anchor graphs to establish the correlations between pixel samples and local regions.

Note that the resulting anchor graphs exhibit multiple scales due to the different number of anchors for each modality. Determining the number of anchors to be equal to the number of superpixels can naturally reflect the specific spatial textures of each modality. Therefore, these anchor graphs can leverage the spatial information sufficiently and explore the cluster structure fully for the corresponding modality.

D. One-Pass Clustering With MSSAGF

After obtaining multiple anchor graphs at different scales, we target fusing the local discriminative structure of each modality into a global consensus structure and achieving the final clustering results of RSIs. However, existing anchor graph fusion methods obtain the global anchor graph by a linear combination of all graphs. This is element by element and requires all anchor graphs to have the same scale. Thus, it is inherently impracticable to be applied to anchor graphs at different scales. Besides, these methods ignore the interpretability of clustering results.

To tackle the above-emerged problems, we view anchor graphs as newly discriminative representations of original data based on anchors, effectively reflecting the near neighbor's relationships between pixels and anchors. When the distance between anchor-based representations of two pixels is small, they are likely to have the same neighboring relationships with the same anchors and have a high confidence belonging to the same cluster. Along this line, we consider the multiple-scale anchor graph fusion problem as a distance-based graph learning problem. Concretely, we integrate global anchor graph learning and dynamic anchor learning of intramodality to obtain an optimal unified anchor graph across multiple local anchor graphs. Furthermore, we introduce an adaptive weighting manner to control the proportion of each modality, acting on the unified global anchor graph learning. Thus, the new model can be expressed as follows:

$$\begin{aligned} &\min_{\mathbf{S},\mathbf{A}^v,a_v} \sum_{v=1}^{\ell} \frac{1}{a_v} \sum_{i=1}^{n} \sum_{j=1}^{\overline{m}} \left\| \mathbf{Z}^v_{i,:} - \mathbf{A}^v_{j,:} \right\|_F^2 \mathbf{S}_{i,j} + \alpha \|\mathbf{S}\|_F^2 \\ &\text{s.t., } \mathbf{S}_{i,:}\mathbf{1} = 1, \quad \mathbf{S}_{i,j} \geq 0, \ \mathbf{a1} = 1, \ a_v \geq 0 \end{aligned} \tag{4}$$

where $\mathbf{A}^v \in \mathbb{R}^{\overline{m} \times m_v}$ indicates dynamic anchors of the vth modality and $\mathbf{Z}^v \in \mathbb{R}^{n \times m_v}$ denotes prebuilt local spectral–spatial anchor graphs of different scales. $\mathbf{S} \in \mathbb{R}^{n \times \overline{m}}$ is the unified global anchor graph. It can absorb the discriminant group structure from multiple anchor graphs at different scales to achieve the optimal clustering structure. The constraint $\mathbf{S}_{i,:}\mathbf{1} = 1, \mathbf{S}_{i,j} \geq 0$ guarantees that the similarity value is nonnegative and the sum is 1 between each sample and all anchors. The constraint $\mathbf{a1} = 1, a_v \geq 0$ offers an interpretation for the contributions of different views and allows $(1/a_v)$ to serve as an adaptive weighting parameter for the vth modality.

Many anchor graph-based methods require postprocessing to obtain the final clustering indicators [15], [37]. However, separate processes may lead to suboptimal indicators, and the graph partitioning process lacks the interpretability of clustering results. To address this issue, we apply orthogonal k-means to the learned global anchor graph to enhance discriminability

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and directly generate clustering indicators. Mathematically, our final MSSAGF model is formulated as follows:

$$\begin{aligned}\min_{\mathbf{S},\mathbf{A}^v,\mathbf{F},\mathbf{G},a_v} &\sum_{v=1}^{\ell}\frac{1}{a_v}\sum_{i=1}^{n}\sum_{j=1}^{\overline{m}}\left\|\mathbf{Z}_{i,:}^v-\mathbf{A}_{j,:}^v\right\|_F^2\mathbf{S}_{i,j}+\alpha\|\mathbf{S}\|_F^2\\ &+\beta\left\|\mathbf{S}-\mathbf{G}\mathbf{F}^T\right\|_F^2\\ \text{s.t., }&\mathbf{S}_{i,:}\mathbf{1}=1,\quad \mathbf{S}_{i,j}\geq 0,\ \mathbf{F}^T\mathbf{F}=\mathbf{I}_k\\ &\mathbf{G}_{i,j}\in\{0,1\},\quad \mathbf{G}_{i,:}\mathbf{1}=1,\ \mathbf{a1}=1,\ a_v\geq 0\end{aligned}\tag{5}$$

where the first two terms indicate the unified global anchor graph learning and the third term represents the clustering partitioning process. $\mathbf{G}\in\mathbb{R}^{n\times k}$ is the clustering indicator matrix with $\mathbf{G}_{i,j}=1$ if the ith instance is assigned to the jth cluster and 0 otherwise. $\mathbf{F}\in\mathbb{R}^{\overline{m}\times k}$ denotes the centroid matrix (k indicates the number of categories). The orthogonal constraint on $\mathbf{F}$ is imposed to enhance discriminability. β is a tradeoff parameter. To the best of our knowledge, this is the first time to fuse anchor graphs at different scales for clustering large-scale RSIs, where multiscale anchor graph fusion and clustering partitioning enhance each other to recover global group information and achieve better clustering performance.

E. Solution

To solve the optimization problem in (5), we introduce the alternating minimizing strategy to optimize each variable with the others being fixed.

1) Updating $\mathbf{F}$*:* When $\mathbf{S},\mathbf{G}$ are fixed, the optimization for $\mathbf{F}$ can be formulated as

$$\min_{\mathbf{F}}\left\|\mathbf{S}-\mathbf{G}\mathbf{F}^T\right\|_F^2,\quad \text{s.t., }\ \mathbf{F}^T\mathbf{F}=\mathbf{I}_k.\tag{6}$$

Expanding the Frobenius norm by trace, (6) can be transformed as

$$\max_{\mathbf{F}}\operatorname{Tr}\left(\mathbf{F}^T\mathbf{S}^T\mathbf{G}\right),\quad \text{s.t., }\ \mathbf{F}^T\mathbf{F}=\mathbf{I}_k.\tag{7}$$

According to [20], the solution of $\mathbf{F}$ is $\mathbf{U}_\mathbf{F}\mathbf{V}_\mathbf{F}^\mathbf{T}$, supposing the singular value decomposition (SVD) result of $\mathbf{S}^T\mathbf{G}$ is $\mathbf{U}_F\Sigma_F\mathbf{V}_F^T$.

2) Updating $\mathbf{G}$*:* When $\mathbf{S},\mathbf{F}$ are fixed, $\mathbf{G}$ can be updated by solving the following problem:

$$\min_{\mathbf{G}}\left\|\mathbf{S}-\mathbf{G}\mathbf{F}^T\right\|_F^2,\quad \text{s.t., }\ \mathbf{G}_{i,j}\in\{0,1\},\ \mathbf{G}_{i,:}\mathbf{1}=1.\tag{8}$$

Since there is only one nonzero entry (i.e., 1) in each row of the indicator matrix $\mathbf{G}$, the assignment problem can be solved independently for each object as follows:

$$\min_{\mathbf{G}_{i,:}}\left\|\mathbf{S}_{i,:}-\mathbf{G}_{i,:}\mathbf{F}^T\right\|_F^2,\quad \text{s.t., }\ \mathbf{G}_{i,j}\in\{0,1\},\ \mathbf{G}_{i,:}\mathbf{1}=1.\tag{9}$$

According to [47], (9) can be solved by seeking the optimal index j^* to minimize the distance between the object and the cluster centroid in the following assignment problem:

$$j^*=\arg\min_j\left\|\mathbf{S}_{i,:}-\mathbf{F}_{:,j}^T\right\|_F^2.\tag{10}$$

3) Updating $\mathbf{S}$*:* When $\mathbf{F},\mathbf{G},a_v,\mathbf{A}^v$ are fixed, the optimization of $\mathbf{S}$ as follows:

$$\begin{aligned}\min_{\mathbf{S}}&\sum_{v=1}^{\ell}\frac{1}{a_v}\sum_{i=1}^{n}\sum_{j=1}^{\overline{m}}\left\|\mathbf{Z}_{i,:}^v-\mathbf{A}_{j,:}^v\right\|_F^2\mathbf{S}_{i,j}+\alpha\|\mathbf{S}\|_F^2+\beta\left\|\mathbf{S}-\mathbf{G}\mathbf{F}^T\right\|_F^2\\ \text{s.t., }&\mathbf{S}_{i,:}\mathbf{1}=1,\quad \mathbf{S}_{i,j}\geq 0.\end{aligned}\tag{11}$$

Denoting $\mathbf{P}=\mathbf{G}\mathbf{F}^\mathbf{T}$, and $\mathbf{D}_{i,j}=\sum_{v=1}^{\ell}(1/a_v)\|\mathbf{Z}_{i,:}^v-\mathbf{A}_{j,:}^v\|_F^2$, (11) can be solved by updating each row vector $\mathbf{S}_{i,:}$ independently as follows:

$$\begin{aligned}\min_{\mathbf{S}_{i,:}}&\frac{1}{2}\left\|\mathbf{S}_{i,:}+\frac{\mathbf{D}_{i,:}}{2\alpha}\right\|_F^2+\frac{\beta}{2\alpha}\left\|\mathbf{S}_{i,:}-\mathbf{P}_{i,:}\right\|_F^2\\ \text{s.t., }&\mathbf{S}_{i,:}\mathbf{1}=1,\quad \mathbf{S}_{i,j}\geq 0.\end{aligned}\tag{12}$$

As proved, (12) can be solved with a closed-form solution [18] as follows:

$$\mathbf{S}_{i,j}=\left(\frac{\frac{-D_{i,j}}{2}+\alpha P_{i,j}+\beta\eta}{\beta+\alpha}\right)_+\tag{13}$$

where $\eta=(1/c)(1+(\alpha/\beta)+\sum_{t=1}^{c}(D_{i,t}/2\beta))$ and $D_{i,1},\ldots,D_{i,\overline{m}}$ are ordered from small to large. Besides, the tuning of parameter α can be transformed to the setting of the number of neighboring anchors c, which is an integer and has explicit meaning.

4) Updating a_v*:* When $\mathbf{S},\mathbf{A}$ are fixed, the optimization of $\{a_v\}_{v=1}^{\ell}$ becomes

$$\begin{aligned}\min_{\mathbf{a}}&\sum_{v=1}^{\ell}\frac{1}{a_v}\sum_{i=1}^{n}\sum_{j=1}^{\overline{m}}\left\|\mathbf{Z}_{i,:}^v-\mathbf{A}_{j,:}^v\right\|_F^2\mathbf{S}_{i,j}\\ \text{s.t., }&\mathbf{a1}=1,\quad a_v\geq 0.\end{aligned}\tag{14}$$

Denoting $h_v=(\sum_{i=1}^{n}\sum_{j=1}^{\overline{m}}\|\mathbf{Z}_{i,:}^v-\mathbf{A}_{j,:}^v\|_F^2\mathbf{S}_{i,j})^{1/2}$, (14) can be rewritten as

$$\min_{\{a_v\}_{v=1}^{\ell}}\sum_{v=1}^{\ell}\frac{(h_v)^2}{a_v},\quad \text{s.t., }\ \sum_{v=1}^{\ell}a_v=1,\ a_v\geq 0.\tag{15}$$

This is a Cauchy–Schwarz's inequality problem, we can update $\mathbf{a}_v$ by

$$a_v=\frac{h_v}{\sum_{v=1}^{\ell}h_v}.\tag{16}$$

5) Updating $\mathbf{A}^v$*:* When $\mathbf{S},\mathbf{a}_v$ are fixed, we can update $\mathbf{A}^v$ by solving the following problem:

$$\min_{\mathbf{A}^v}\sum_{i=1}^{n}\sum_{j=1}^{\overline{m}}\left\|\mathbf{Z}_{i,:}^v-\mathbf{A}_{j,:}^v\right\|_F^2\mathbf{S}_{i,j}.\tag{17}$$

The above equation can be rewritten as

$$\min_{\mathbf{A}^v}\operatorname{Tr}\left(\mathbf{Z}^{v^T}\mathbf{W}\mathbf{Z}^v-2\mathbf{Z}^{v^T}\mathbf{S}\mathbf{A}^v+\mathbf{A}^{v^T}\mathbf{Q}\mathbf{A}^v\right)\tag{18}$$

where $\mathbf{W}=\operatorname{diag}(\sum_{j=1}^{\overline{m}}\mathbf{S}_{i,j})$ and $\mathbf{Q}=\operatorname{diag}(\sum_{i=1}^{n}\mathbf{S}_{i,j})$. Taking the derivation of $\mathbf{A}_v$ in (18) and setting it to zero, we can get

$$\mathbf{A}^v=\mathbf{Q}^{-1}\mathbf{S}^T\mathbf{Z}^v\tag{19}$$

with optimizing the multiple subproblems separately, the entire optimization procedure of solving problem (5) is summarized in Algorithm 1.

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Algorithm 1 MSSAGF

Input: Multimodal RSI $\{\mathcal{X}^v \in \mathbb{R}^{h\times w\times b_v}\}_{v=1}^{\ell}$ and the number of classes k, denoising parameter λ, the number of dynamical anchors $\overline{m}$, clustering parameters β, the number of neighbors c.

Output: Clustering indicator $\mathbf{G}$.

Prebuilt $\mathbf{Z}^v$ at different scales, Initialize $\mathbf{A}^v$, Initialize $\mathbf{S}$, Initialize a_v with $\frac{1}{\ell}$.
while not converge **do**
  Update $\mathbf{F}$ by solving Eq. (7);
  Update $\mathbf{G}$ via the optimal index by Eq. (10);
  Update $\mathbf{S}$ by solving Eq. (12);
  Update a_v by solving Eq. (16);
  Update $\mathbf{A}^v$ by solving Eq. (19);
end while
Output the final clustering indicator matrix $\mathbf{G}$.

F. Complexity Analysis

1) Computational Complexity: The computational expense of the proposed MSSAGF contains the cost of the optimization of each subproblem. Specifically, to solve $\mathbf{F}$, it takes $\mathcal{O}(\overline{m}k^2)$ and $\mathcal{O}(n\overline{m}k)$ for SVD and matrix multiplication. To update $\mathbf{G}$, it costs $\mathcal{O}(n\overline{m}k)$ to execute assignment. To learn $\mathbf{S}$, it needs $\mathcal{O}(n\overline{m}\xi)$ to get the closed-form solution, where $\xi = \sum_{v=1}^{\ell} m_v$ is the summation of the number of superpixels. To update a_v, it takes $\mathcal{O}(1)$. To solve $\mathbf{A}^v$, it needs $\mathcal{O}(n\overline{m}m_v)$ for matrix multiplication. Besides, we construct an initial anchor graph $\mathbf{Z}^v$, taking $\mathcal{O}(nm_vd_v)$. Suppose the optimization algorithm 1 stop after t iterations, the overall computational complexity is $\mathcal{O}((\overline{m}k^2 + n\overline{m}k + n\overline{m}\xi)t + \sum_{v=1}^{\ell} nm_vd_v)$. Note that $k \ll n, m_v \ll n, \overline{m} \ll n, \xi \ll n, d_v \ll n$, the computational complexity of MSSAGF is nearly linear to the number of samples $\mathcal{O}(n)$.

2) Space Complexity: During the optimization procedure, MSSAGF needs to store $\{\mathbf{Z}^v, \mathbf{S}, \mathbf{G}, \mathbf{A}^v, \mathbf{F}\}$, their space costs are $\mathbf{Z}^v \in \mathbb{R}^{n\times m_v}$, $\mathbf{S} \in \mathbb{R}^{n\times\overline{m}}$, $\mathbf{G} \in \mathbb{R}^{n\times k}$, $\mathbf{A}^v \in \mathbb{R}^{\overline{m}\times m_v}$, and $\mathbf{F} \in \mathbb{R}^{\overline{m}\times k}$, respectively. Thus, the space complexity is $\mathcal{O}(n(\xi+\overline{m}+k)+\overline{m}\xi+\overline{m}k)$. This is comparable to the storage cost of multimodal RSIs.

IV. Experimental Results and Analysis

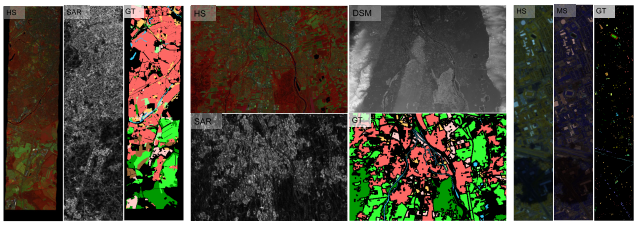
To evaluate our method, we perform extensive experiments on three multimodal RSI datasets to present performance quantitatively and qualitatively. We first introduce the datasets used, the comparison methods, and the experimental setup. Next, we investigate the clustering performance of our model, including running time, parameter sensitivity, and the ablation studies, to demonstrate the effectiveness of our approach.

A. Experimental Details

1) Datasets: HS-synthetic aperture radar (SAR) Berlin: The Berlin scene comprises a size of 1723 × 476 pixels and contains eight ground-truth (GT) objects. This dataset includes an HS image with 244 spectral bands and an SAR image with four channels. HS-SAR-DSM Augsburg: The Augsburg scene is composed of three different sources. They are an HS image with 180 spectral bands, an SAR image with four features, and a DSM image with one channel, respectively. The dataset has a size of 332 × 485 pixels and includes seven GT objects. HS-multispectral (MS) Houston 2013: The Houston scene contains two different sources. They are an HS image with 144 spectral bands and an MS image with eight channels. This dataset contains 15 GT objects and has a size of 349 × 1905 pixels. For visual understanding, we showcase the false-color visualization and GT distribution of these three datasets in Fig. 3. Besides, Table I summarizes the detailed land-cover types of these datasets.

TABLE I
Description of the Berlin, Augsburg, and Houston 2013 RSI Datasets

Berlin		**Houston**	
Class Name	#Samples	Class Name	#Sample
Forest	54954	Healthy grass	1251
Residential Area	268642	Stressed grass	1254
Industrial Area	19566	Synthetic grass	697
Low Plants	59282	Trees	1244
Soil	17426	Soil	1242
Allotment	13305	Water	325
Commercial Area	24824	Residual	1268
Water	6672	Commercial	1244
Total	464671	Road	1252
Augsburg		Highway	1227
Forest	13507	Railway	1235
Residential Area	30329	Parking Lot1	1233
Industrial Area	3851	Parking Lot2	469
Low Plants	26857	Tennis Court	428
Allotment	575	Running Track	660
Commercial Area	1645		
Water	1530		
Total	78294	Total	15029



(a) (b) (c)

Fig. 3. False-color visualization of three multimodal RSI datasets: HS-SAR Berlin, HS-SAR-DSM Augsburg, and HS-MS Houston 2013. (a) Berlin. (c) Augsburg. (d) Houston.

2) Compared Methods: We compare our method with ten competitors. These methods can be roughly into three categories.

1) *Single-modal methods for RSIs:* FSCAG [37], SGCNR [38], SGLSC [35], NCSC [36], and MS^{2}A-Net [26].
2) *Multimodal methods for GIs:* FPMVS-CAG [21], AMvC [22], and MFLVC [23].
3) *Multimodal methods for RSIs:* MDC [13], TMPCC [14], and AMKSC [15]. The introduction of these methods can be found in Sections I and II. For a clear comparison, we summarize these methods in Table II. Note that we concatenated all the different modalities along the bands as input for the single-modal methods.

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TABLE II
COMPARISON OF OUR METHOD WITH THE OTHERS MENTIONED

Methods	Cluster ways	C1	C2	C3	C4	C5
FSCAG	SVD and k-means	✓	✓	✗	✗	✓
SGCNR	continuous indicators	✓	✓	✗	✗	✗
SGLSC	EVD and k-means	✓	✓	✗	✗	✓
NCSC	EVD and k-means	✓	✓	✗	✗	✗
MSA-Net	k-means	✓	✗	✗	✗	✓
FPMVS-CAG	SVD and k-means	✓	✓	✓	✗	✗
AMvC	SVD and k-means	✓	✓	✓	✗	✗
MFLVC	softmax operation	✓	✗	✓	✗	✗
MDC	k-means	✓	✗	✓	✗	✓
TMPCC	softmax operation	✓	✗	✓	✗	✓
ours	discrete indicators	✓	✓	✓	✓	✓

C1– 5 DENOTE WHETHER THE CORRESPONDING METHOD IS SCALABLE, USE GRAPH STRUCTURE, CAN SUITABLE FOR MULTIMODAL DATASETS, CAN USE DIFFERENT SCALES GRAPHS, AND CAN EXPLOIT SPATIAL INFORMATION, RESPECTIVELY. SVD MEANS SINGULAR VALUE DECOMPOSITION, AND EVD MEANS EIGENVALUE DECOMPOSITION

TABLE III
INFORMATION OF IMPORTANT PARAMETERS FOR OUR METHODS. k DENOTES THE NUMBER OF CATEGORIES

Parameters		Berlin	Augsburg	Houston
Setting	λ	10^2	10^4	10^{-2}
	β	10^{-6}	10^{-5}	10^{-1}
	c	20	25	10
	$\overline{m}$	$4k$	$6k$	$5k$
Range	λ	$[10^{-4}, 10^{-3}, ..., 10^3, 10^4]$		
	β	$[10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}]$		
	c	$[10, 15, 20, 25]$		
	$\overline{m}$	$[3k, 4k, 5k, 6k, 7k]$		

3) Experimental Setup: Following [2], the number of superpixels is adaptively determined based on the texture of each RSI by

$$m_v = \delta \frac{n_{\text{edge}}^v}{n} \tag{20}$$

where n_{edge}^v denotes the number of nonzero elements in the detected edge image of the vth modality, n indicates the number of pixels, and δ is an adjustable parameter and default as 2000. Moreover, the local spectral–spatial anchor graph at different scales of each modality is constructed as introduced in Section III-C. Additionally, some related parameters of our method should be tuned to adapt to the characteristics of the data. In our experiments, we use grid search to obtain the best parameter settings. Table III presents the ranges of the parameters and their optimal settings. Our method achieves acceptable performance across a wide range of parameters (see Section IV-C), effectively demonstrating the practicality of our approach. Table III exhibits different optimal parameter settings across datasets due to their varying properties. In practice, it is straightforward to identify feasible parameters that make our model effective by slightly tuning the parameter $\overline{m}$ and c. The deep learning-based compared methods are implemented in Python 3.7.13, and the other competitors are implemented in MATLAB. For the implementation of our proposed method, the edge detection for RSI segmentation is based on the Image Processing Toolbox in MATLAB. All experiments are run on a desktop computer with a 2.3 GHz Intel[1] Xeon[1] Gold 5119 CPU Processor and 128 GB RAM.

For a fair comparison, we use the released source codes of these compared methods in our experiments. Following the settings in the original articles, we tune their parameters to the optimum and report the average results of ten repeated trials. To quantify the clustering performance, six widely used metrics, that is, clustering accuracy (ACC), normalized mutual information (NMI), purity and adjusted rand index (ARI), Kappa coefficient (Kappa), and F-score, are used to evaluate the clustering performance. Higher values indicate better clustering performance.

B. Comparison Results With State-of-the-Arts

1) Clustering Performance: Table IV presents the quantitative clustering results of the eleven methods on three datasets. The best results are marked in bold, meanwhile "N/A" means the results are unavailable on the Augsburg dataset since MDC is only suitable for two modalities. Based on Table IV, we can draw the following conclusions.

1) Spatial information is vital for pixel-wise recognition. Some single-modal methods for RSIs, such as SGLSC, can achieve comparable performance with multimodal methods for GIs by considering spatial information. Although achieving multimodal scenarios, multimodal methods devised for GIs still present some limitations in addressing RSI data. For instance, FPMVS-CAG shows the discouraging results. The reason may be that the spatial information in RSIs is ignored.
2) Multimodal data has great potential in exploring the group information, but existing multimodal methods for RSIs only show comparable performance with that of single-modal methods. We suppose that this may be because these methods focus on extracting similar features without adequately mining the structural information hidden in data and the spatial context of each modality.
3) Our method presents the best performance in most circumstances against all compared methods. Specifically, our method exceeds the second-best method by 13.24%, 19.65%, and 2.4% increments in terms of ACC on the Berlin, Augsburg, and Houston datasets, respectively. Besides, our method achieves 7.03%, 19.44%, and 1.95% improvement over the second-best methods by averaging six metrics on three datasets, respectively. All methods show poor performance on the Houston dataset because this dataset contains various real-world challenging scenarios.

To demonstrate the clustering performance of the above methods intuitively, we provide a visual comparison through clustering maps displayed in Figs. 4–6. As can be observed, the superiority of the proposed MSSAGF is aligned with the quantitative results.

2) Running Time: Table V presents the running time of all multimodal clustering methods on Augsburg, Berlin,

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TABLE IV

CLUSTERING RESULTS OF DIFFERENT METHODS ON THREE DATASETS. THE 1ST BEST RESULTS ARE MARKED IN BOLD. "N/A" MEANS THE RESULTS ARE UNAVAILABLE

Datasets	Metric	Unimodal Methods					Multimodal methods for GI			Multimodal methods for RSI			
		FSCAG	SGCNR	SGLSC	NCSC	MSA^2-Net	FPMVS-CAG	AMvC	MFLVC	MDC	TMPCC	AMKSC	Ours
Berlin	ACC	0.5080	0.4892	0.5348	0.2801	0.5711	0.3282	0.5108	0.4166	0.5083	0.4456	0.4426	**0.7035**
	NMI	0.3134	0.2785	0.3368	0.2491	0.3193	0.1872	0.1850	0.3119	0.3187	**0.4163**	0.2897	0.3896
	Purity	0.7199	0.6574	0.7252	0.6775	0.7117	0.5781	0.6403	0.6836	0.6617	**0.7701**	0.6939	0.7238
	ARI	0.2467	0.2251	0.3204	0.0672	0.3335	0.0076	0.2159	0.2143	0.2621	0.2526	0.1927	**0.4242**
	Kappa	0.3331	0.3135	0.3580	0.1617	0.3877	0.1570	0.2188	0.2641	0.3034	0.3430	0.3052	**0.4593**
	F-score	0.3874	0.3504	0.3491	0.2033	0.3719	0.1587	0.2508	0.3327	0.2525	0.3748	0.3869	**0.4164**
	Average	0.4181	0.3857	0.4374	0.2732	0.4492	0.2361	0.3369	0.3705	0.3845	0.4337	0.3851	**0.5195**
Augsburg	ACC	0.4462	0.3622	0.5399	0.3263	0.5702	0.4059	0.5009	0.5468	N/A	0.6269	0.5738	**0.8234**
	NMI	0.2895	0.2267	0.4477	0.1752	0.3252	0.1577	0.2721	0.3884	N/A	0.4600	0.4430	**0.5857**
	Purity	0.6088	0.5749	0.7903	0.5358	0.6518	0.5035	0.6046	0.7466	N/A	0.7559	0.7769	**0.8256**
	ARI	0.2231	0.1387	0.3963	0.0867	0.3545	0.1012	0.2317	0.3191	N/A	0.4029	0.3742	**0.6110**
	Kappa	0.2879	0.2138	0.4334	0.2161	0.4409	0.1697	0.3211	0.4390	N/A	0.4996	0.4530	**0.7355**
	F-score	0.3311	0.2955	0.3801	0.2369	0.3392	0.2021	0.2885	0.3694	N/A	0.3264	0.3921	**0.3945**
	Average	0.3644	0.3020	0.4979	0.2628	0.4470	0.2567	0.3698	0.4682	N/A	0.5119	0.5021	**0.6626**
Houston	ACC	0.4714	0.4416	0.3618	0.3085	0.3879	0.2906	0.3601	0.4256	0.4310	0.4211	0.4815	**0.4954**
	NMI	0.4978	0.4726	0.3891	0.4012	0.4115	0.4148	0.3854	0.5156	0.4791	0.4411	**0.5337**	0.5023
	Purity	0.4970	0.4732	0.3892	0.3481	0.4106	0.2955	0.3714	0.4648	0.4665	0.4211	0.5028	**0.5051**
	ARI	0.3058	0.2643	0.1909	0.1757	0.2465	0.1871	0.1857	0.2908	0.2782	0.2626	**0.3321**	0.3099
	Kappa	0.4298	0.3975	0.3104	0.2577	0.3383	0.2297	0.3061	0.3847	0.3869	0.3752	0.4401	**0.4551**
	F-score	0.3915	0.3746	0.3338	0.2682	0.3005	0.1984	0.2983	0.3827	0.3699	0.3061	0.3921	**0.4426**
	Average	0.4322	0.4039	0.3292	0.2932	0.3492	0.2694	0.3178	0.4107	0.4019	0.3712	0.4470	**0.4517**

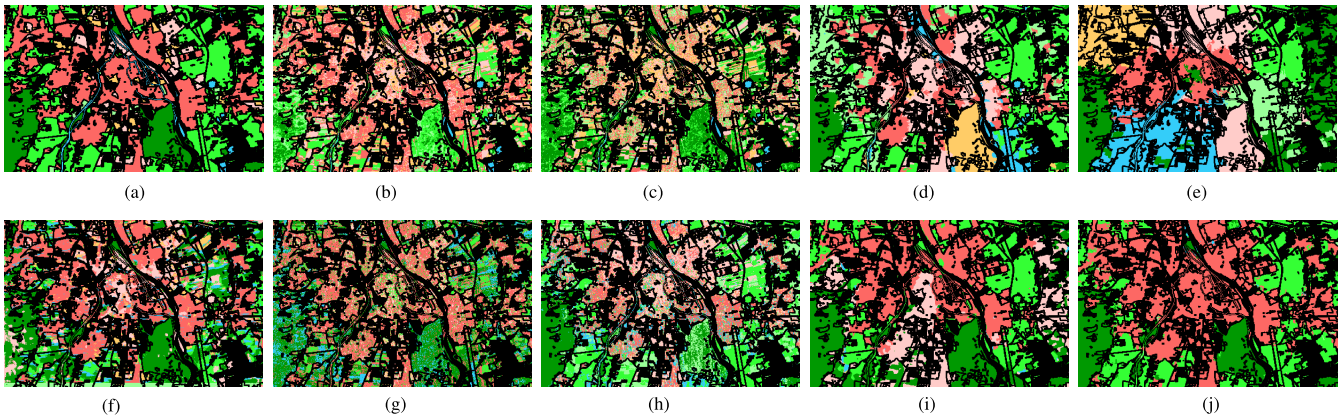


Fig. 4. Clustering maps and GT for the Augsburg dataset obtained by different clustering methods. (a) GT. (b) FSCAG. (c) SGCNR. (d) SGLSC. (e) NCSC. (f) MS^{2}A-Net. (g) AMvC. (h) MFLVC. (i) TMPCC. (j) Ours.

TABLE V

TIME COMPLEXITY AND RUNNING TIME (t/s) COMPARISONS OF DIFFERENT MULTIMODAL CLUSTERING METHODS ON THREE RSI DATASETS (MEASURED BY SECONDS)

Methods	Time Complexity	Running Time		
		Augsburg	Berlin	Houston
MFLVC	$\mathcal{O}(nwt)$	165.91	310.27	161.26
MDC	$\mathcal{O}(nwt)$	N/A	433.52	230.04
AMKSC	$\mathcal{O}(nm^2t)$	65.97	484.86	465.22
AMvC	$\mathcal{O}(nwt)$	4028.18	17509.36	17042.98
TMPCC	$\mathcal{O}(nwt)$	6342.55	21127.67	18653.45
FPMVS-CAG$*$	$\mathcal{O}(nm^3dt)$	314.92	2023.89	1640.02
Ours$*$	$\mathcal{O}(nm^2t)$	238.68	1495.70	1539.10

n, m, w, d, t: the number of pixels, anchors, parameters, bands, and iterations. $*$ indicates the non-deep learning methods, the rest are deep learning methods.

and Houston datasets, respectively. We can observe that our proposed method demonstrates higher efficiency than FPMVS-CAG, AMvC, and TMPCC methods. Although MFLVC and MDC achieve higher efficiency than our proposed method, they are deep learning-based methods, requiring complicated optimization processes. Our method exceeds MFLVC by 14.9%, 19.44%, and 4.1% increments in terms of average performance on three datasets and exceeds MDC by 13.8%, 4.98% increments.

C. Parameter Sensitivity Analysis

In our method, the parameter λ in (2) is crucial for denoising, while the number of dynamical anchors $\overline{m}$ and the number of neighbors c are related to the global anchor graph S. Besides, the parameter β in (5) is vital for obtaining clustering results. To investigate their sensitivities, we report the performance variance by varying their values in Fig. 7. Specifically, λ is tuned in the range of $[10^{-4} : 10 : 10^4]$, β is

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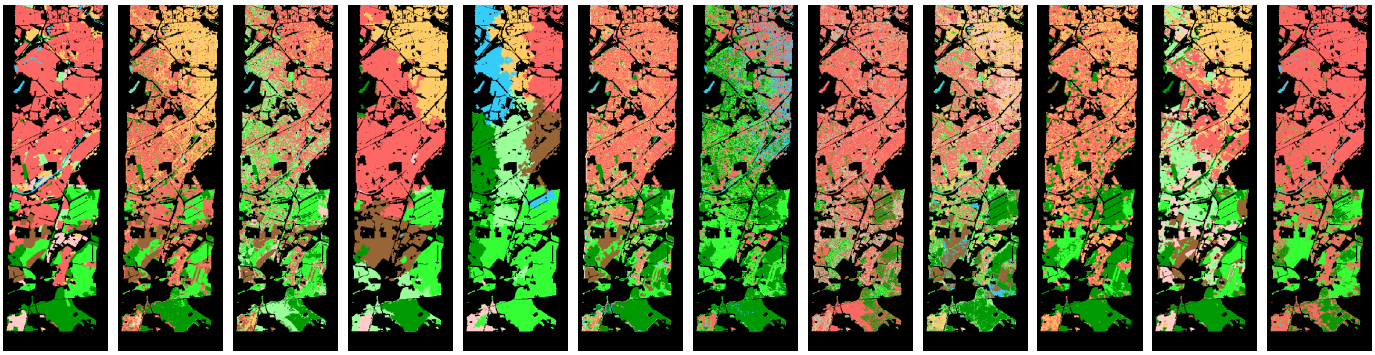


Fig. 5. Clustering maps and GT for the Berlin dataset obtained by different clustering methods. (a) GT. (b) FSCAG. (c) SGCNR. (d) SGLSC. (e) NCSC. (f) MS^{2}A-Net. (g) FPMVS. (h) AMvC. (i) MFLVC. (j) MDC. (k) TMPCC. (l) Ours.

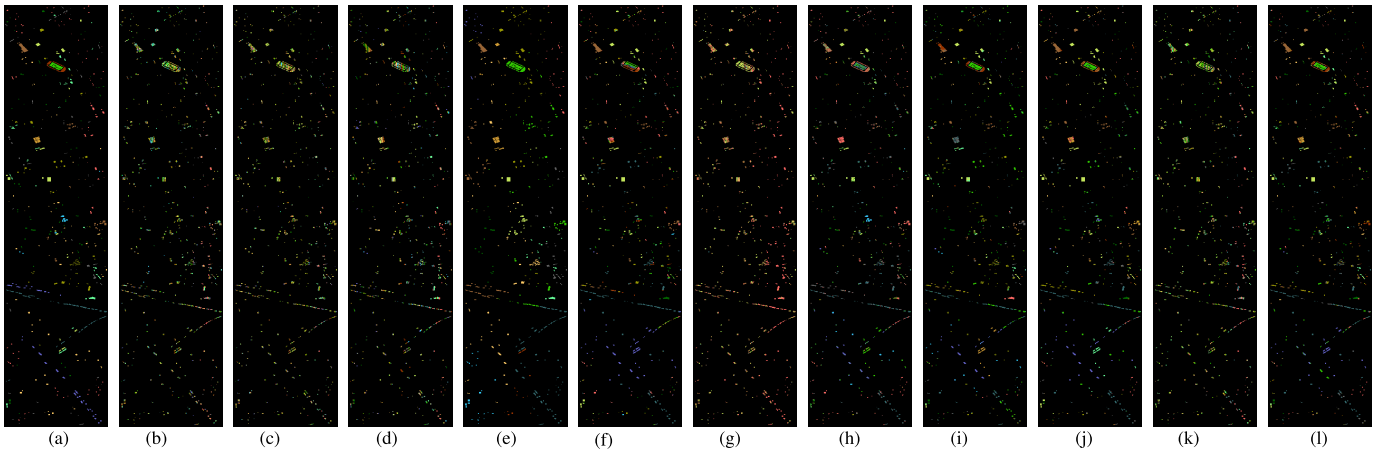


Fig. 6. Clustering maps and GT for the Houston dataset obtained by different clustering methods. (a) GT. (b) FSCAG. (c) SGCNR. (d) SGLSC. (e) NCSC. (f) MS^{2}A-Net. (g) FPMVS. (h) AMvC. (i) MFLVC. (j) MDC. (k) TMPCC. (l) Ours.

TABLE VI
ABLATION STUDY OF OUR MSSAGF ON RSI DATASETS

Setting	Components		Berlin			Augsburg			Houston		
	SNR	MS	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity
S0	✘	✘	0.6619	0.3439	0.6650	0.6443	0.4871	0.7777	0.4698	0.4918	0.4889
S1	✘	✔	0.6912	0.3086	0.6912	0.7379	0.5098	0.7739	0.4838	0.4819	0.4924
S2	✔	✘	0.6898	0.3206	0.6926	0.7278	0.4686	0.7478	0.4861	0.4883	0.4944
ours	✔	✔	**0.7035**	**0.3896**	**0.7238**	**0.8234**	**0.5857**	**0.8256**	**0.4954**	**0.5023**	**0.5051**

SNR: Superpixel-Based Neighborhood Recover; MS: Multi-scale Strategy.

tuned in the range of $[10^{-6} : 10 : 10^{-1}]$, $\overline{m}$ is turned from $3k$ to $7k$ with step k, and c is tuned in the range of $[10, 15, 20, 25]$. Although different values produce different clustering results, our method demonstrates relatively stable performance across a wide range of parameters. In detail, for parameter β, our method is relatively insensitive, and tuning it only leads to slight fluctuations, reflecting the robustness of the orthogonal k-means classifier. For parameter λ, we usually set larger values to enhance the spatial smoothness of small superpixels, such as in the Augsburg dataset. This is because pixels within small superpixels are usually more homogeneous. For the number of anchors $\overline{m}$ and the number of neighbors c, a small range of values is sufficient to obtain optimal performance. Typically, $\overline{m} \in \{4k, 5k, 6k\}$ and $c \in \{10, 15, 20, 25\}$ are recommended as candidates for various datasets. Therefore, grid searching can easily find the optimal values of λ, $\overline{m}$, and c. This further demonstrates the practicality of our method.

D. Ablation Study

To investigate the effect of each component, we conduct an ablation study to show the degraded variants of MSSAGF by removing different components. For the degraded variants, we utilize the notations "✔" and "✘" to indicate their configuration, containing a certain component or not. As can be observed in Table VI, the variant "S0" indicates removing both superpixel-based neighborhood recovery and multiscale strategy. The variant "S1" denotes removing superpixel-based neighborhood recovery. We use the local reconstruction approach of [25] as a substitution to execute

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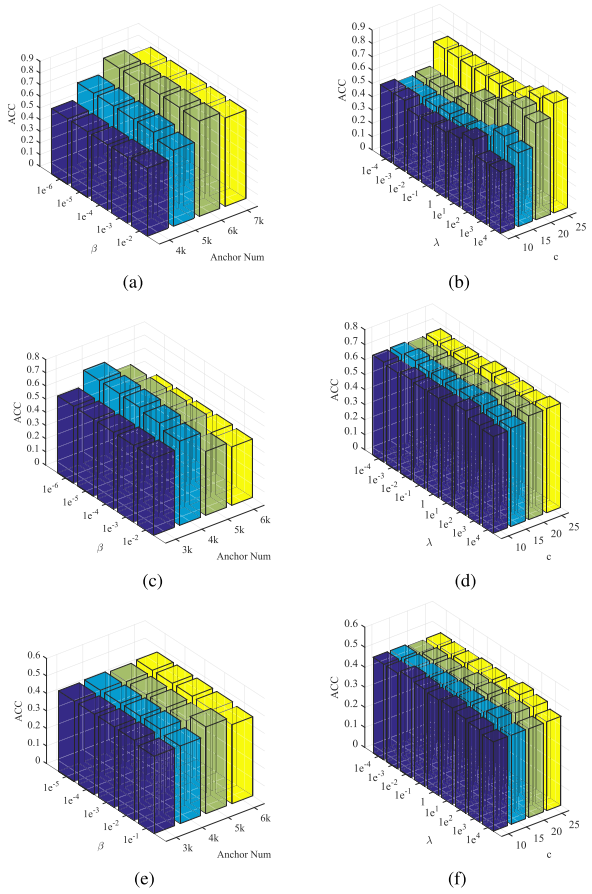


Fig. 7. Sensitivity analysis. (Left) The column presents the performance with varying β and $\overline{m}$. (Right) The column presents the performance with varying λ and c. (a) Augsburg. (b) Augsburg. (c) Berlin. (d) Berlin. (e) Houston. (f) Houston.

RSI denoising. The variant "S2" indicates removing the multiscale strategy. We concatenate multiple modalities along bands and segment them to generate the superpixels, resulting same number of anchors for each modality. Under the multiscale strategy, "S1" outperforms "S0" in most circumstances. It verifies that the multiscale strategy can make full use of the diverse spatial information in multiple modalities. Comparing "S2" and "S0," we can observe that using the superpixel-based neighborhood recovery approach can enhance the clustering accuracy. Consequently, by introducing both superpixel-based neighborhood recovery and multiscale strategy, our proposed method shows the best performance.

V. Conclusion

In this article, we proposed a novel scalable multimodal RSI clustering method termed MSSAGF. It can fully exploit the spatial information of each modality based on different spatial textures. Based on the diverse texture information, multiple RSIs were separately segmented into different numbers of superpixels to reflect the different spatial interdependencies of land covers. Then, we proposed a superpixel-based nonlinear neighborhood recovery approach to denoise each superpixel. Meanwhile, the average spectrum of each superpixel was used as a spatial-aware anchor to construct a spectral–spatial anchor graph for each modality. For different scales of spectral–spatial anchor graphs, we proposed an adaptive fusion model to fuse them into a unified global anchor graph and learn clustering indicators simultaneously. Moreover, an alternating optimization approach was designed to solve this objective function. It can directly provide clustering results with linear time complexity. Extensive experiments have demonstrated the superior performance of our MSSAGF in comparison with many state-of-the-art clustering methods. Our future work is to develop parameter-free methods for automatic parameter selection. Moreover, continual learning is a promising direction for multimodal RSI clustering.

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